Samarth Sharma

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Profile

Postgraduate Diploma in Artificial Intelligence graduate (University of Manchester) with research experience in compiler-aware performance modelling of deep learning accelerators, and hands-on experience building machine learning systems, from data pipelines and LSTM forecasting to NLP and sentiment analysis. Seeking to deepen my research experience in a computational research lab.

Education

The University of Manchester Sep 2024 – Aug 2025
– Postgraduate Diploma in Artificial Intelligence

The University of East London Sep 2021 – Jun 2024
– Bachelor of Science (Hons.) in Computer Science — **Distinction**

Research Projects

Compiler-Aware Performance Modeling of NVDLA Mar 2025 – Aug 2025
Postgraduate Diploma Dissertation — University of Manchester
Compared NVIDIA NVDLA's native Caffe-based compiler with ONNC using LeNet-5, AlexNet, and SqueezeNet. ONNC reduced memory footprint by **12–22%** and instruction counts by up to **27%** through operator fusion and buffer reuse.
Technologies: Python, ONNC, Caffe, NVDLA, ONNX

Relation Extraction using Deep Learning (SemEval-2010) Jan 2025 – Mar 2025
Built and evaluated SVM and multi-head attention BiLSTM models for semantic relation classification, reaching up to **82.9% F1-score** using TF-IDF, POS tagging, dependency parsing, and GloVe embeddings.
Technologies: Python, PyTorch, scikit-learn, spaCy, Gensim

Smart Voting System Using Blockchain Technology Submitted May 2024
BSc Dissertation — University of East London
Designed and built a decentralised voting prototype on Ethereum, backed by a literature review of blockchain voting (including Estonia's e-voting) and a user survey of 18 respondents.
Technologies: Solidity, Ethereum, Truffle, React, Node.js, MongoDB, MetaMask

Selected Projects

FlowCast: AI Traffic Prediction Platform Internmo — Individual project
Traffic prediction system covering data ingestion and validation, cleaning and merging, classical machine learning, an LSTM model, and a Streamlit dashboard.

Northgate AI Stock Predictor Internmo
Predicts stock prices using machine learning, deep learning, and financial news sentiment analysis, with portfolio optimisation and Buy, Hold, or Sell recommendations based on risk and expected returns.

Vantara Customer Intelligence Platform Internmo — In progress
Analytics platform that predicts customer churn, estimates customer lifetime value, and analyses purchase behaviour to support retention and personalised marketing.

Work Experience

AI Engineering Intern Internmo — Sep 2026 – Present
– Building three industry-level AI projects (see Selected Projects).

Systems and Network Administrator Assistant University of East London — Feb 2023 – Feb 2024
– Monitored system health and troubleshot networks across university IT infrastructure supporting multiple users and services.
– Deployed and configured software on Linux and Windows, documented incidents and maintenance for audits, and provided first-line support and onboarding.

Micro-Intern — Manchester Micro Internship Programme University of Manchester — Apr 2025 – May 2025
– Completed a three-week consultancy project with an industry client in a multidisciplinary team, presenting findings to stakeholders and delivering a report with actionable recommendations.

Microsoft Embrace Mentoring Programme — Mentee Microsoft — Sep 2022 – Dec 2022
– Completed a career-readiness programme and built a personal development plan from my technical strengths and skill gaps.

Skills

Programming	Python, Java, C/C++, Bash, Solidity
Machine Learning / AI	PyTorch, scikit-learn, NumPy, Pandas, LSTM and attention models, time-series forecasting, NLP (spaCy, Gensim), sentiment analysis
Web & APIs	REST APIs, Node.js, Express, React, Streamlit
Systems & Tools	Git, Docker, Linux, Wireshark
Databases	SQL (MySQL), MongoDB
Concepts	Data Structures and Algorithms, Software Testing, SDLC, Agile

Additional

– Member of a Japanese language society, learning Japanese
– Trinity College London, Grade 8 Keyboard (2013 – 2021)
– Certificate of Recognition: Mental Wealth – Professional Life 2